Back-of-the-Envelope Estimation

which numbers to ask for first

Watch the 3-min video

In one line: before you estimate, ask for daily and monthly users, actions per user (reads and writes), object size, retention, peak vs average traffic and where users are. If a number is missing, state a round assumption out loud.

1 Why estimate first

"Design a URL shortener." Before you draw a single box you need to know: is this a hobby project with a **thousand links**, or a global service creating **a hundred million links a day**? The architecture is completely different. One gets a single small server and a database; the other needs caching, replication and sharding.

So the first step of any estimate is **asking for the right numbers**. The goal is not a precise answer. It is the right **order of magnitude**: 10, 1,000 or 100,000 requests per second, gigabytes or petabytes. That is what decides the design.

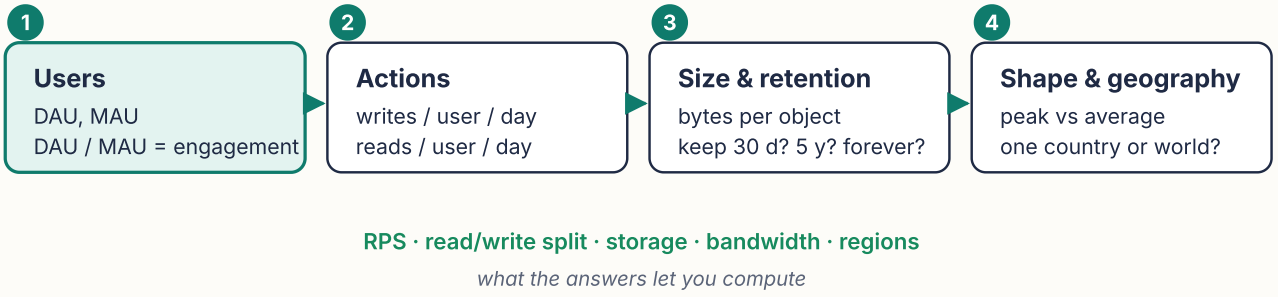


Figure 1. Four groups of questions, asked in this order. Each answer feeds a number you will need later.

2 Users: DAU and MAU

DAU means daily active users: people who use the product on a given day. **MAU** means monthly active users. Interviewers often give you one of them.

The ratio **DAU / MAU** tells you how engaged users are. A social app might see half of its monthly users every day (50%). A tool people open once a week sits far lower. If you know MAU and the ratio, you get DAU, and DAU drives traffic.

3 Actions, sizes and retention

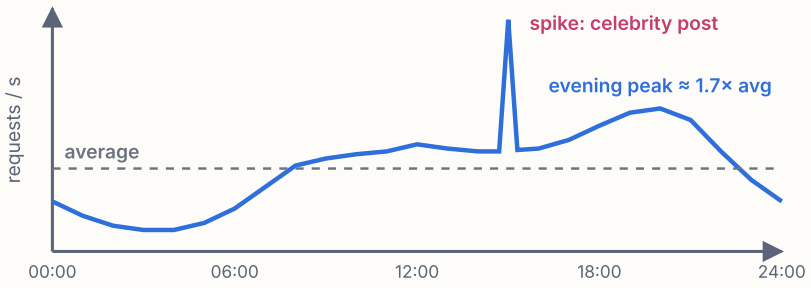
Actions per user per day. How many times a day does a user **write** something (post, upload, create a link)? How many times do they **read** (view a feed, open a link)? From that you get the **read-to-write ratio**. Most systems are read-heavy, often **10:1**, sometimes **100:1**. The ratio tells you whether to invest in caches and replicas or in write throughput.

Size of each object. A few hundred bytes for a database row, a few megabytes for a photo, gigabytes for a video. **Retention:** 30 days, 5 years, or forever?

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storage = size per object × objects per day × days retained
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4 Shape and geography

Is traffic flat, or does it spike: at lunchtime, on Black Friday, or when a celebrity posts? **Peak is often 2–10× the average**, and you size for the peak. And where are the users: one country, or worldwide? That decides how many regions you run, how you replicate data, and whether you need a CDN.



Ask about shape
is it flat, or does it spike?
peak ≈ 2–10× average

Ask about geography
one country or worldwide?
→ regions, replication, CDN

Figure 2. An average hides the evening peak and the sudden spike. Ask for the shape, not just the daily total.

5 Worked example: the URL shortener

Ask the questions, get the answers, then compute. Use 86,400 seconds per day (about 10^5 for quick math).

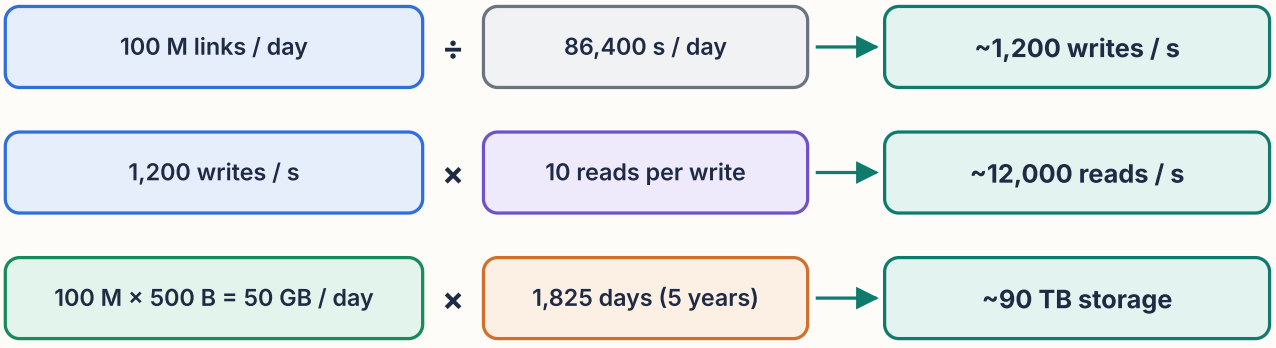


Figure 3. Four answers give three numbers: write rate, read rate and storage.

WORKED EXAMPLE

- **Writes:** 100 M links ÷ 86,400 s ≈ **1,157**, call it **~1,200 writes/s**.
- **Reads:** 10 redirects per link → 10 × 1,157 ≈ **~12,000 reads/s**. Read-heavy: cache the hot short codes.
- **Storage:** 100 M × 500 B = 50 GB/day; × 1,825 days ≈ **91 TB**, about **90 TB** before replication. That is about 183 billion links: too much for one comfortable database node, so plan to shard.

6 When the interviewer gives no number

Don't freeze. **State an assumption out loud**: “Let's assume 100 million links a day.” Round it so the math is easy, and **write it down** where both of you can see it. The interviewer can correct you, and your reasoning stays easy to follow. A wrong but stated assumption is fine; a silent one is not.

COMMON MISTAKES

- Drawing the architecture before knowing whether it is 10 or 10,000 requests per second.
- Chasing exact arithmetic (1,157.4 writes/s) instead of the order of magnitude (~1,000).
- Forgetting retention, so storage is computed for one day instead of five years.
- Sizing for the daily average and ignoring the 2–10× peak.
- Making assumptions silently, so the interviewer can't follow or correct them.

7 Key terms

DAU / MAU: daily / monthly active users

Retention: how long data must be kept

Back-of-the-envelope: quick, rounded estimate to pick a design

Engagement: DAU ÷ MAU, e.g. 50%

Peak factor: peak ÷ average traffic, often 2–10×

Assumption: a stated, rounded number used when none is given

Read:write ratio: reads per write, often 10:1 to 100:1

Order of magnitude: the power of ten: 10^3 vs 10^5

RPS: requests per second

8 Interview questions

Q1. Which numbers do you ask for before estimating? BASIC

DAU and MAU, actions per user per day split into reads and writes, the read-to-write ratio, the size of each object, retention, peak vs average traffic, and where users are located.

Q2. What does the DAU/MAU ratio tell you? BASIC

Engagement: the share of monthly users active on a given day. 100 M DAU out of 200 M MAU is 50%, typical of a sticky social app; a weekly tool might be closer to 15–20%.

Q3. How do you compute storage? BASIC

Size per object × objects per day × days retained. 100 M records/day × 500 B × 1,825 days ≈ 91 TB, before replication (× 3 copies ≈ 270 TB of disk).

Q4. Estimate the URL shortener: 100 M links/day, 10:1 reads, 5 years, 500 B each. SCENARIO

~1,200 writes/s, ~12,000 reads/s and ~90 TB over five years. Read-heavy, so cache redirects; 90 TB means the link table will be sharded.

Q5. The interviewer won't give you a DAU. What do you do? SCENARIO

State a round assumption out loud (“let's assume 10 million DAU”), write it down, and carry on. Offer to redo the math if they prefer another number; the method matters more than the input.

Q6. Why ask where users are located? SCENARIO

It decides regions, replication and CDN. Worldwide users far from one region see 150 ms+ round trips, so you may serve reads from several regions and static content from a CDN.

Q7. Your estimate says 1,157 writes per second. How do you present it? TRICKY

As ~1,000 (or ~1,200) writes per second. Estimates are for orders of magnitude; false precision hides the fact that the inputs were rough guesses.

Q8. Why does retention matter more than daily volume for storage? TRICKY

Storage grows with every day you keep. 50 GB/day is trivial for one day but ~91 TB over five years; with “forever” it never stops growing, so you also need archiving or tiering.

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